IMAGE ENCRYPTION AND DECRYPTION

Using the 5D tangent-hyperbolic memristive chaotic model

SUMMARY: The supplied parrot image (250 × 334 RGB pixels) was encrypted with a deterministic keystream generated from the 5D model. Decryption with the same keystream recovered every decoded RGB pixel exactly.

1. Objective and research basis

The objective was to demonstrate image encryption and decryption using chaotic data from the five-dimensional tangent-hyperbolic memristive model in Qureshi et al. (2024). The supplied JPEG test image was converted to an 8-bit RGB array before processing. The encrypted and decrypted outputs were saved as lossless PNG files.

The paper introduces an Amalgamated Image Encryption (AIE) approach that uses chaotic 3D flow data together with a second plain image as key material. Only one test image was supplied for this run, so the experiment below uses the chaotic keystream alone; it is a reproducible XOR-based demonstration, not a full two-image AIE reproduction.

2. 5D chaotic model

The first equation is to place γ outside the hyperbolic tangent:

$$\mathbf{dx_1/dt = a(x_2 - x_1) + 0.25x_3 + g(\delta + \gamma \tanh(x_5))x_3}$$

State equation	Right-hand side
dx_2/dt	$b\,x_1 - x_1x_3 + c\,x_2 - x_4$
dx_3/dt	$d\,x_1x_2 - e\,x_3$
dx_4/dt	x_1
dx_5/dt	x_3

Parameter	Value	Initial condition	Value
a	78	$x_1(0)$	0
b	−16	$x_2(0)$	0
c	58	$x_3(0)$	8
d	35	$x_4(0)$	6
e	15	$x_5(0)$	0
g, δ, γ	1, 1, 0.5		

Numerical method: classical fourth-order Runge–Kutta (RK4), step h = 0.001. This is the integer-order case α = 1. The notebook does not implement the paper's Atangana–Baleanu–Caputo fractional scheme for α = 0.99.

3. Image encryption and decryption flow

Step	What the code does
1. Input	Read the supplied JPEG and convert it to RGB uint8 array (height 334, width 250).
2. Generate chaotic states	Run the 5D system using RK4; discard 10,000 transient steps.
3. Form keystream	Use x_1, x_2 and x_3; sample every 10 integration steps and convert each value to a byte using floor($\lvert x \rvert \times 10^8$) mod 256.
4. Encrypt	XOR the RGB image with the three-channel chaotic keystream.
5. Decrypt	XOR the encrypted image with the exact same keystream.
6. Verify and save	Compare the decrypted array to the original RGB array; save encrypted/decrypted images as PNG.

4. Encryption equations

$$\mathbf{C = P \oplus K}$$

$$\mathbf{\hat{P} = C \oplus K = P}$$

P is the original RGB image, K is the chaotic byte keystream, C is the encrypted image, and $\hat{P}$ is the recovered image. XOR is its own inverse, so using the same key stream a second time restores the original bytes.

5. Actual outputs from the supplied test image

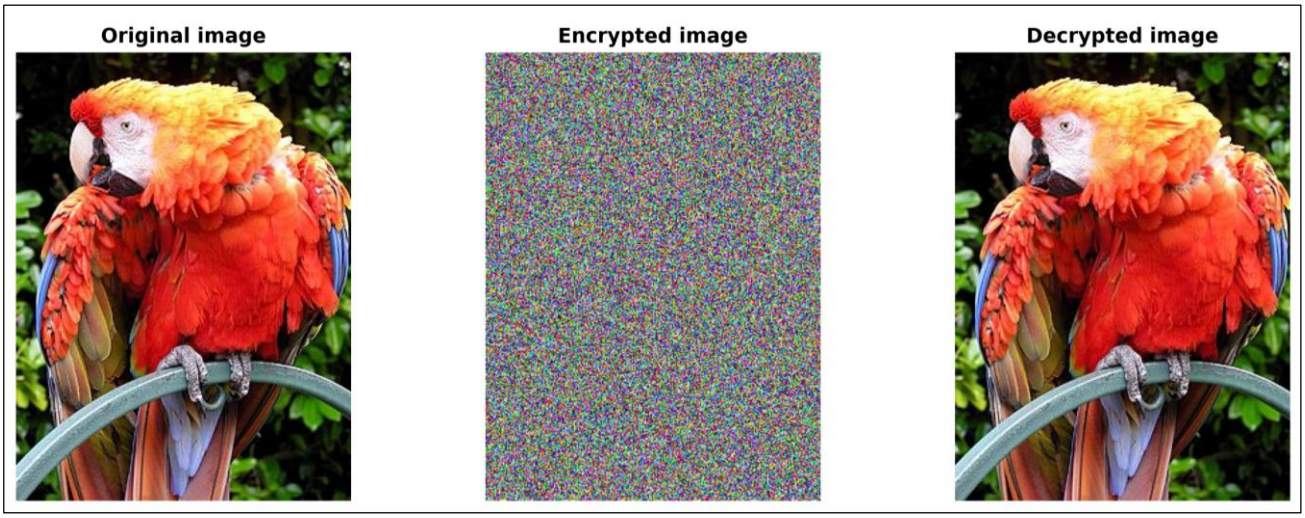


Figure 1. Original parrot image, encrypted result, and decrypted result. The decrypted output visually matches the original.

The encrypted image appears noise-like and the original image is recovered after decryption. The exact array comparison (not only visual inspection) returned True. The encrypted image is saved as PNG to avoid adding compression artefacts.

6. Test results

Test	Measured result	Interpretation
Image dimensions	250 × 334 × 3	RGB; 83,500 pixels
Correct-key recovery	Exact = True	All decoded RGB pixel values match
MSE after decryption	0	No pixel error
PSNR after decryption	∞ dB	Expected when MSE = 0
Entropy R (original → encrypted)	7.6052 → 7.9974 bits	Maximum is 8 bits/channel
Entropy G (original → encrypted)	7.6077 → 7.9974 bits	Encrypted values close to maximum
Entropy B (original → encrypted)	7.0631 → 7.9979 bits	Encrypted values close to maximum
Horizontal correlation R	0.9459 → 0.0024	Reduced strongly
Horizontal correlation G	0.9197 → −0.0363	Reduced strongly
Horizontal correlation B	0.9034 → −0.0328	Reduced strongly
Wrong initial condition	Exact = False; PSNR = 6.93 dB	$x_1(0)$ changed by 10^{-10}
One-pixel differential NPCR	0.0011976%	One changed pixel out of 83,500
One-pixel differential UACI	0.00000470%	Small because the XOR scheme has no diffusion

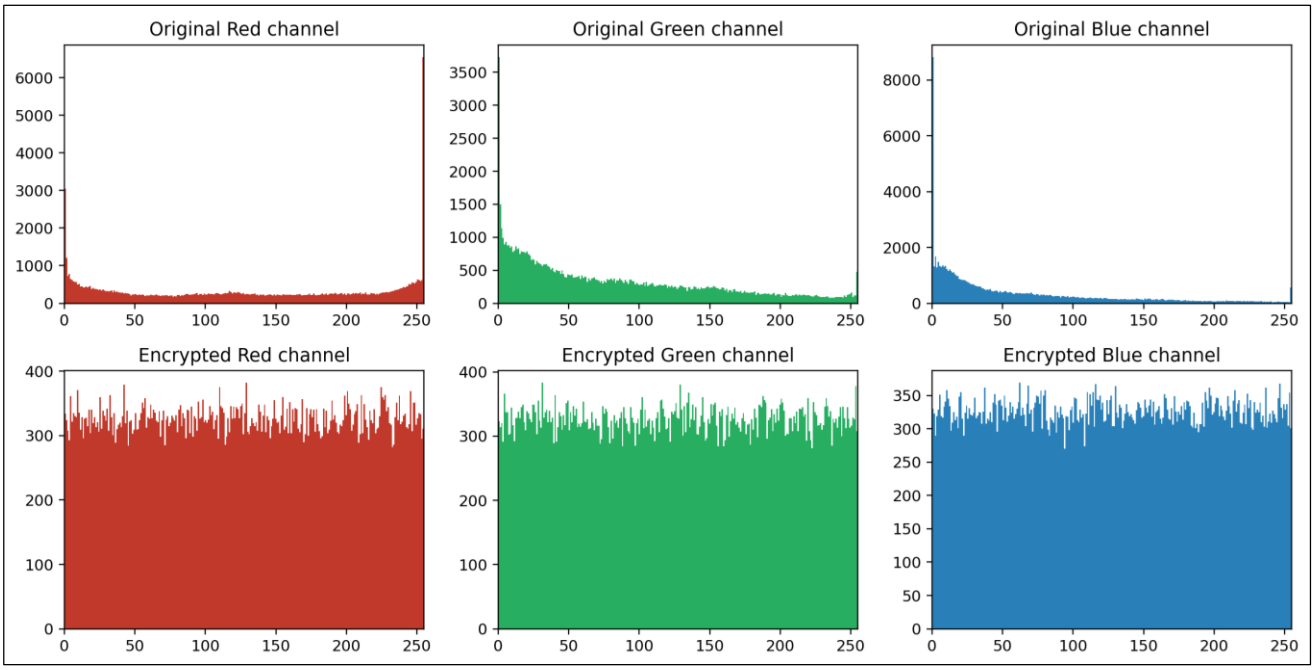


Figure 2. Original and encrypted RGB-channel histograms. The encrypted-channel histograms are substantially more uniform for this test image.

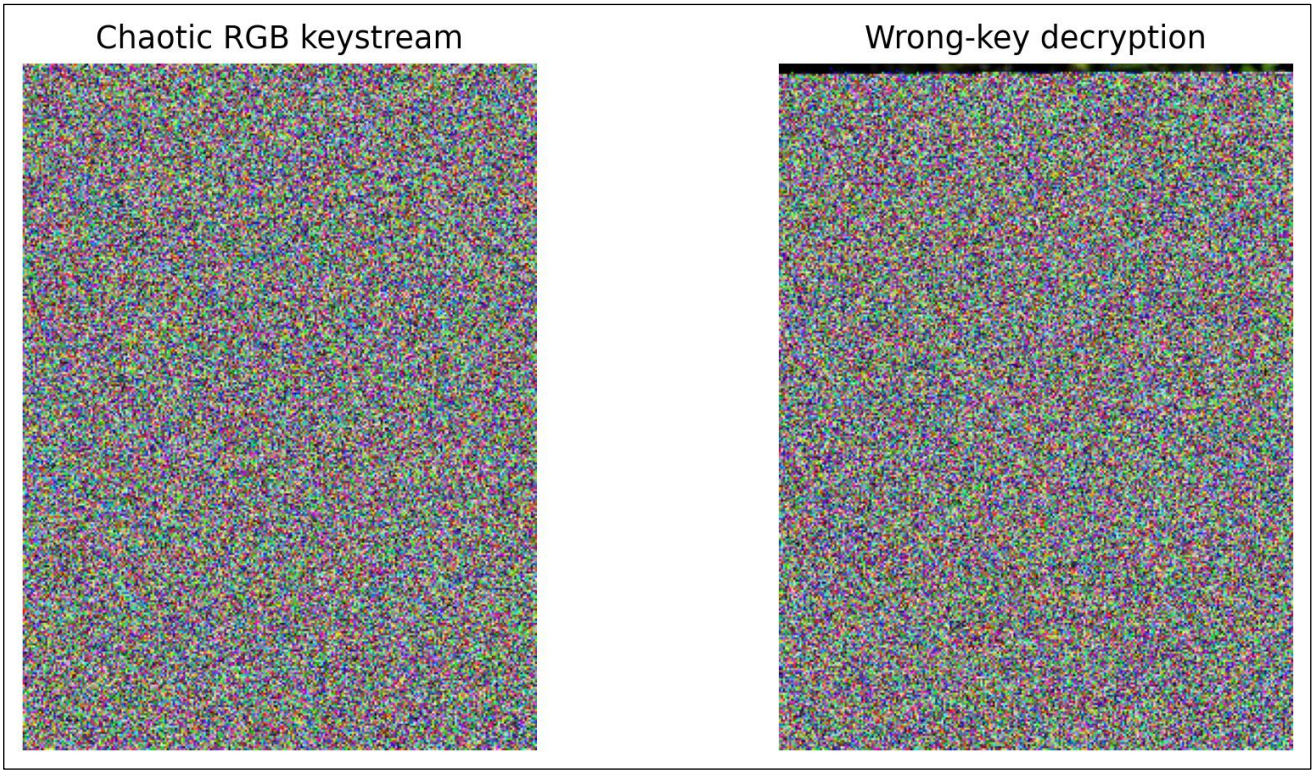


Figure 3. Generated chaotic RGB keystream and result of decryption with a slightly incorrect initial condition.

7. How to interpret the results

- Correct decryption: MSE = 0 and infinite PSNR are the mathematically expected results when the same XOR keystream is applied twice. This confirms reversibility for the decoded RGB array.
- Entropy and histogram: encrypted channel entropy is approximately 7.997 bits out of 8, and the histograms are much flatter than those of the input image. These are positive statistical observations, but they do not prove cryptographic security.
- Correlation: the sampled horizontal neighbour correlation drops close to zero after encryption. This indicates reduced local pixel correlation in this experiment.
- Wrong key: changing $x_1(0)$ by 10^{-10} prevents exact recovery and produces a low-PSNR result. This is an example of sensitivity to the chosen initial condition, not a complete cryptanalysis.
- NPCR/UACI: these values are low because the implemented cipher is only XOR with a deterministic mask. A one-pixel input change affects only that pixel in the ciphertext. Do not compare this differential test directly with the different values reported in the paper's Table 2.

8. Important limitations

- Only one image was supplied. No second plain-image key was used, although the paper's AIE description uses chaotic flow data and a second image as key material.
- The paper's ABC fractional-order $\alpha = 0.99$ numerical scheme is not implemented here; this run uses RK4 for $\alpha = 1$ only.
- The published parameter values and initial conditions are public in the paper; they demonstrate reproducibility and should not be presented as a secret cryptographic key.
- The byte conversion $\text{floor}(|x| \times 10^8) \bmod 256$ is a code-specific design choice, not a formula prescribed by the paper.
- The input was JPEG. Exact recovery refers to the decoded 8-bit RGB pixels. It does not reproduce the original compressed JPEG file byte-for-byte.
- The XOR-only design has no diffusion and reuses a deterministic keystream for a fixed key. It should be treated as an experimental prototype, not as proven secure encryption for real confidential data.

9. Conclusion

The 5D model generated a three-channel chaotic keystream that transformed the supplied parrot image into a noise-like encrypted image. Using the same keystream, the decryption recovered the original decoded RGB array exactly (MSE = 0, PSNR = ∞). Entropy, histogram and correlation measurements were favourable for this test image, while the one-pixel NPCR/UACI experiment exposed the expected lack of diffusion in the XOR-only construction. The result demonstrates a successful reversible image-encryption pipeline, but it is not a complete reproduction of the paper's two-image AIE or its fractional-order $\alpha = 0.99$ simulation.

Code flow

Image → RGB array → 5D model / RK4 → x_1, x_2, x_3 → byte keystream → XOR encryption → encrypted PNG → XOR decryption → exact RGB comparison

Reference

Qureshi, M. A., Khan, N. A., Raza, S., & Iqbal, S. M. Z. (2024). "Cap like trajectories in 5D chaotic tangent hyperbolic memristive model: fractional calculus and encryption." Physica Scripta, 99, 075238. https://doi.org/10.1088/1402-4896/ad5476

Github link:

https://github.com/fizza-14/5D-Chaotic-Memristive-Image-Audio-Video-Encryption-Decryption